



## Bibliography tracking

The following table provides an overview of the references used in the thesis, indicating whether each work has been read and whether it is currently cited in the written material.

Table 1: Tracking of GEO datasets, manifests, and probe-filtering resources

| First author / Source                              | Dataset / Resource title                                                                                                                      | Read | Used | Ref. |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|------|------|------|
| <b>GEO datasets</b>                                |                                                                                                                                               |      |      |      |
| NCBI                                               | GSE69914: DNA methylation profiles in breast tissue samples                                                                                   | Yes  | Yes  | [1]  |
| NCBI                                               | GSE225845: High neighborhood deprivation impacts DNA methylation and gene expression in cancer-related genes                                  | Yes  | Yes  | [2]  |
| NCBI                                               | GSE287331: DNA methylation patterns in breast cancer, benign tissue, and healthy controls                                                     | Yes  | Yes  | [3]  |
| Dennis                                             | DNA methylation patterns in breast cancer, paired benign tissue from ipsilateral and contralateral breast, and healthy controls               | No   | Yes  | [4]  |
| <b>Manifests &amp; annotation resources</b>        |                                                                                                                                               |      |      |      |
| Illumina                                           | Infinium MethylationEPIC v1.0 B5 Manifest File                                                                                                | Yes  | Yes  | [5]  |
| Robinson                                           | IlluminaHumanMethylation450kanno.ilmn12.hg19: annotation package for 450k arrays (Bioconductor)                                               | Yes  | Yes  | [6]  |
| <b>Probe filtering &amp; array-specific issues</b> |                                                                                                                                               |      |      |      |
| Naeem                                              | Reducing the risk of false discovery enabling identification of biologically significant genome-wide methylation status using the HM450 array | Yes  | Yes  | [7]  |
| Chen                                               | Discovery of cross-reactive probes and polymorphic CpGs in the Illumina HumanMethylation450 microarray                                        | Yes  | Yes  | [8]  |
| Pidsley                                            | Critical evaluation of the Illumina MethylationEPIC BeadChip microarray for whole-genome DNA methylation profiling                            | No   | Yes  | [9]  |
| Zhou                                               | Comprehensive characterization, annotation and innovative use of Infinium DNA methylation BeadChip probes                                     | Yes  | Yes  | [10] |
| McCartney                                          | Identification of polymorphic and off-target probe binding sites on the Illumina Infinium MethylationEPIC BeadChip                            | Yes  | Yes  | [11] |

Table 2: Tracking of consulted and cited references for Medical / biological background

| First author                           | Full paper title                                                                                                                    | Read | Cited | Ref. |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------|-------|------|
| <b>Medical / biological background</b> |                                                                                                                                     |      |       |      |
| NIH                                    | Understanding Cancer                                                                                                                | Yes  | No    | [12] |
| NCI                                    | BRCA Gene Changes: Cancer Risk and Genetic Testing Fact Sheet                                                                       | Yes  | Yes   | [13] |
| Azizi                                  | Introduction to Cancer Biology                                                                                                      | Yes  | No    | [14] |
| Menon                                  | Breast Cancer (StatPearls)                                                                                                          | Yes  | No    | [15] |
| Hanahan                                | The Hallmarks of Cancer                                                                                                             | Yes  | No    | [16] |
| Ehrlich                                | DNA Methylation in Cancer: Too Much, but Also Too Little                                                                            | Yes  | Yes   | [17] |
| Bird                                   | DNA Methylation Patterns and Epigenetic Memory                                                                                      | Yes  | Yes   | [18] |
| Russo                                  | Epigenetic Mechanisms of Gene Regulation                                                                                            | Yes  | Yes   | [19] |
| Pina-Sanchez                           | Cancer Biology, Epidemiology, and Treatment in the 21st Century: Current Status and Future Challenges From a Biomedical Perspective | No   | No    | [20] |
| Wilcox                                 | The CpG Landscape of Protein Coding DNA in Vertebrates                                                                              | Yes  | No    | [21] |
| Ji                                     | A comprehensive methylome map of lineage commitment from Hematopoietic Progenitors                                                  | No   | No    | [22] |
| Jenkins                                | Neighborhood Deprivation and DNA Methylation and Expression of Cancer Genes in Breast Tumors                                        | No   | No    | [23] |

Table 3: Tracking of consulted and cited references for Data visualization and exploration

| First author                              | Full paper title                                                                                                                     | Read | Cited | Ref. |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|------|-------|------|
| <b>Data visualization and exploration</b> |                                                                                                                                      |      |       |      |
| Maksimovic                                | SWAN: Subset-quantile Within Array Normalization for Illumina HumanMethylation450 BeadChips                                          | Yes  | Yes   | [24] |
| Benelli                                   | Charting Differentially Methylated Regions in Cancer with Rocker-meth                                                                | Yes  | No    | [25] |
| Seeboth                                   | DNA Methylation Outlier Burden, Health, and Ageing in Generation Scotland and the Lothian Birth Cohorts                              | Yes  | No    | [26] |
| Zhao                                      | A Urine-based DNA Methylation Assay, ProCURE, to Identify Clinically Significant Prostate Cancer                                     | Yes  | No    | [27] |
| Chen                                      | Detecting Early-Warning Signals for Sudden Deterioration of Complex Diseases by Dynamical Network Biomarkers                         | No   | Yes   | [28] |
| West                                      | On Dynamic Network Entropy in Cancer                                                                                                 | No   | Yes   | [29] |
| Liu                                       | Identifying the Critical States and Dynamic Network Biomarkers of Cancers Based on Network Entropy                                   | No   | Yes   | [30] |
| Islam                                     | Integration of DNA Methylation Patterns and Genetic Variation in Human Pediatric Tissues Helps Inform EWAS Design and Interpretation | No   | Yes   | [31] |
| Mansell                                   | Guidance for DNA Methylation Studies: Statistical Insights from the Illumina EPIC Array                                              | No   | Yes   | [32] |
| Chen                                      | mLiftOver: Mapping DNA Methylation Data Across Array Platforms and Genome Builds                                                     | No   | Yes   | [33] |
| Yousefi                                   | Considerations for normalization of DNA methylation data by Illumina 450K BeadChip assay in population studies                       | No   | Yes   | [34] |
| Liu                                       | An evaluation of processing methods for Illumina HumanMethylation450 BeadChip platform                                               | No   | Yes   | [35] |
| Welsh                                     | A systematic evaluation of normalization methods and probe replicability using Infinium EPIC 850K methylation data                   | No   | Yes   | [36] |
| Pidsley                                   | A data-driven approach to preprocessing Illumina 450K methylation array data                                                         | No   | Yes   | [37] |
| Maksimovic                                | A cross-package Bioconductor workflow for analysing methylation array data                                                           | No   | Yes   | [38] |
| Zhang                                     | Critical evaluation of the reliability of DNA methylation probes on the Illumina MethylationEPIC v1.0 BeadChip microarrays           | No   | No    | [39] |
| Liu                                       | An evaluation of processing methods for HumanMethylation450 BeadChip data                                                            | No   | No    | [35] |

Table 4: Tracking of consulted and cited references for GSE69914 Applications

| First author                 | Full paper title                                                                                                              | Read | Cited | Ref. |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------|------|-------|------|
| <b>GSE69914 Applications</b> |                                                                                                                               |      |       |      |
| Chen                         | Discovery of cross-reactive probes and polymorphic CpGs in the Illumina HumanMethylation450 microarray                        | Yes  | No    | [8]  |
| Teschendorff                 | DNA methylation outliers in normal breast tissue identify field defects that are enriched in cancer                           | Yes  | Yes   | [40] |
| Gao                          | The integrative epigenomic-transcriptomic landscape of ER-positive breast cancer                                              | Yes  | No    | [41] |
| Ding                         | Integrative analysis identifies potential DNA methylation biomarkers for pan-cancer diagnosis and prognosis                   | Yes  | No    | [42] |
| Gao                          | DNA methylation patterns in normal tissue correlate more strongly with breast cancer status than CNVs                         | Yes  | No    | [43] |
| Croes                        | Large-scale analysis of DFNA5 methylation reveals its potential as biomarker for breast cancer                                | Yes  | No    | [44] |
| Fan                          | Prediction of disease genes based on stage-specific gene regulatory networks in breast cancer                                 | Yes  | No    | [45] |
| Gao                          | Genome-wide discovery of circulating cell-free DNA methylation signatures for TNBC diagnosis                                  | Yes  | No    | [46] |
| Zhu                          | An improved epigenetic counter to track mitotic age in normal and precancerous tissues                                        | Yes  | No    | [47] |
| Yang                         | An integrative pan-cancer-wide analysis of epigenetic enzymes reveals universal patterns of epigenomic deregulation in cancer | Yes  | No    | [48] |
| Bartlett                     | Inference of tissue relative proportions of breast epithelial subtypes (luminal progenitor, basal, luminal mature)            | Yes  | No    | [49] |
| Teschendorff                 | EPISCORE: Cell-type deconvolution of bulk DNA methylomes using single-cell RNA-seq data                                       | Yes  | No    | [50] |

Table 5: Tracking of consulted and cited references for Data Preparation

| First author            | Full paper title                                                                                                                           | Read | Cited | Ref. |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|------|-------|------|
| <b>Data Preparation</b> |                                                                                                                                            |      |       |      |
| Teschendorff            | A beta-mixture quantile normalization method for correcting probe design bias in Illumina Infinium 450k DNA methylation data               | No   | Yes   | [51] |
| de Almeida              | A multimodal conversational agent for DNA, RNA and protein tasks                                                                           | No   | Yes   | [52] |
| Sokolov                 | Decoding depression: a comprehensive multi-cohort exploration of blood DNA methylation using machine learning and deep learning approaches | No   | Yes   | [53] |

Table 6: Tracking of consulted and cited references for Data Pre-processing &amp; quality control

| First author                                                        | Full paper title                                                                                                                                      | Read | Cited | Ref. |
|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|------|
| <b>Data Pre-processing &amp; quality control MANCANO PAPERSSSSS</b> |                                                                                                                                                       |      |       |      |
| Du                                                                  | Comparison of Beta-value and M-value methods for quantifying methylation levels by microarray analysis                                                | Yes  | Yes   | [54] |
| Cazaly                                                              | Comparison of pre-processing methodologies for Illumina 450k methylation array data in familial analyses                                              | Yes  | No    | [55] |
| Xu                                                                  | The ENmix DNA methylation analysis pipeline for Illumina BeadChip and comparisons with seven other preprocessing pipelines                            | No   | No    | [56] |
| Bizet                                                               | Improving Infinium MethylationEPIC data processing: re-annotation of enhancers and long noncoding RNA genes and benchmarking of normalization methods | No   | No    | [57] |
| Morris                                                              | Analysis pipelines and packages for Infinium HumanMethylation450 BeadChip (450k) data                                                                 | No   | No    | [58] |

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